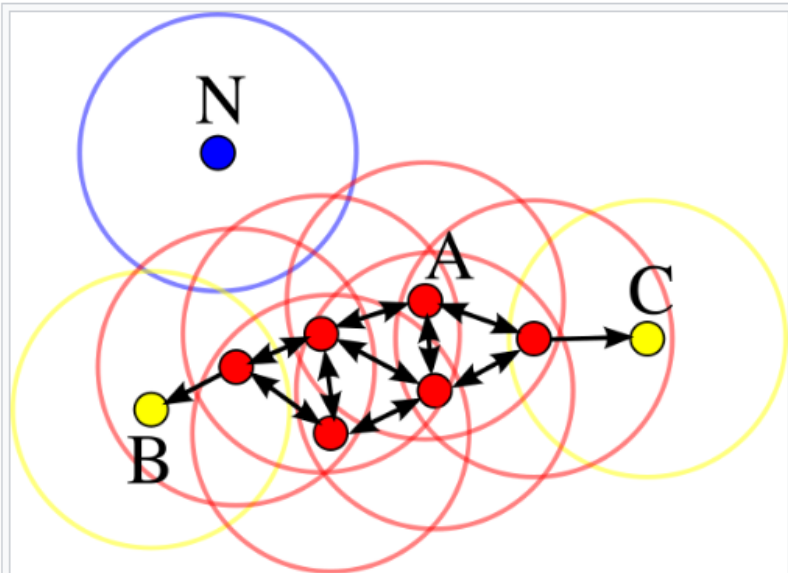


DB SCAN



In this diagram, $\text{minPts} = 4$. Point A and the other red points are core points, because the area surrounding these points in an ϵ radius contain at least 4 points (including the point itself). Because they are all reachable from one another, they form a single cluster. Points B and C are not core points, but are reachable from A (via other core points) and thus belong to the cluster as well. Point N is a noise point that is neither a core point nor directly-reachable.

- core point
- border point
- noise / outlier

It is helpful for non linear clustering.

Hyperparameter

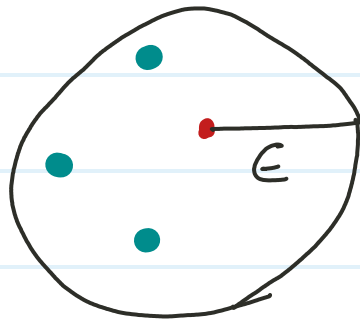
① minpoints = 4

② ϵ = Radius

($\epsilon = \epsilonpsilon$)

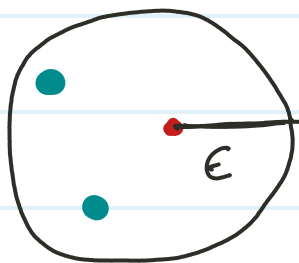
core point -

No. of points within the ϵ should be $\geq$ min point

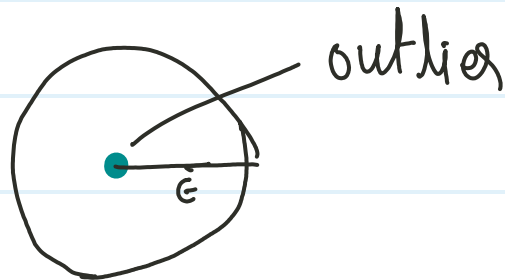


Border point -

No. of data point within the radius will be less than the min point 4.

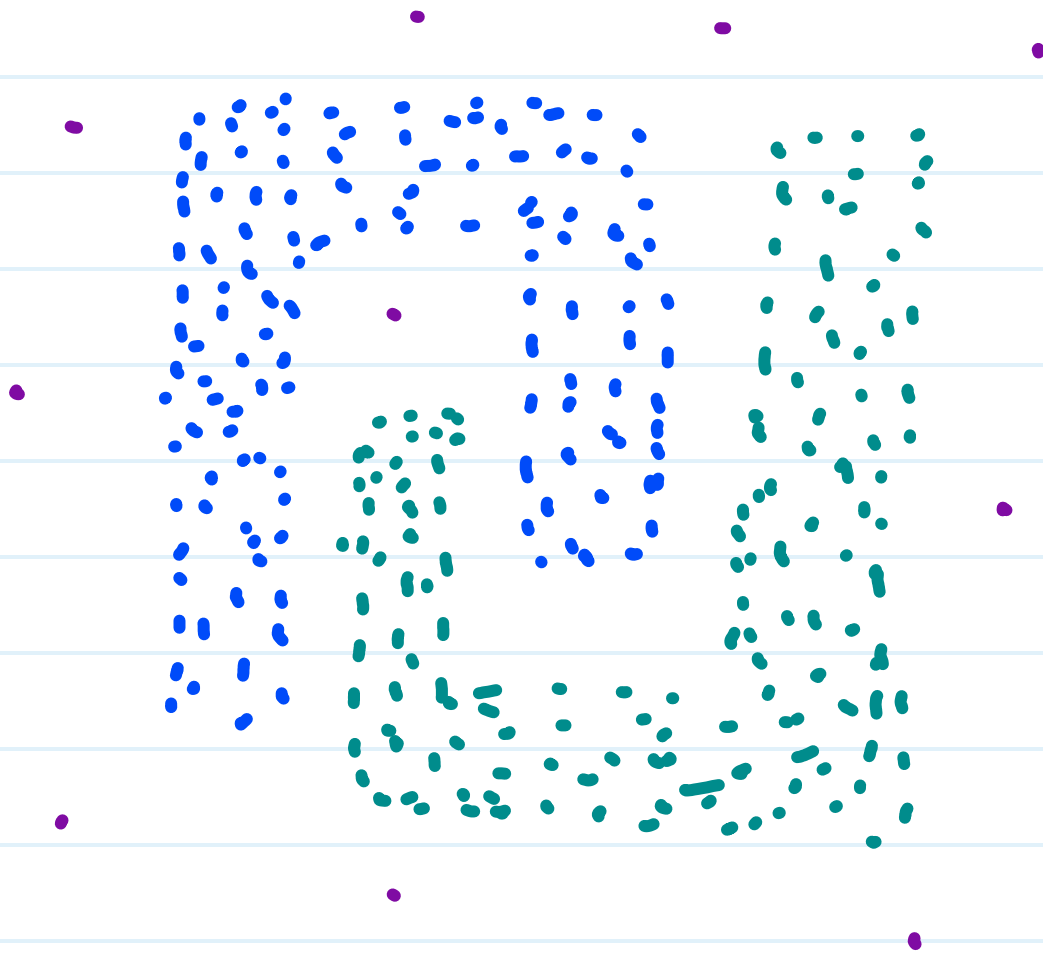


Noise/ outlier - (DBSCAN robust to outlier)
No. of



Some example after applying DBSCAN

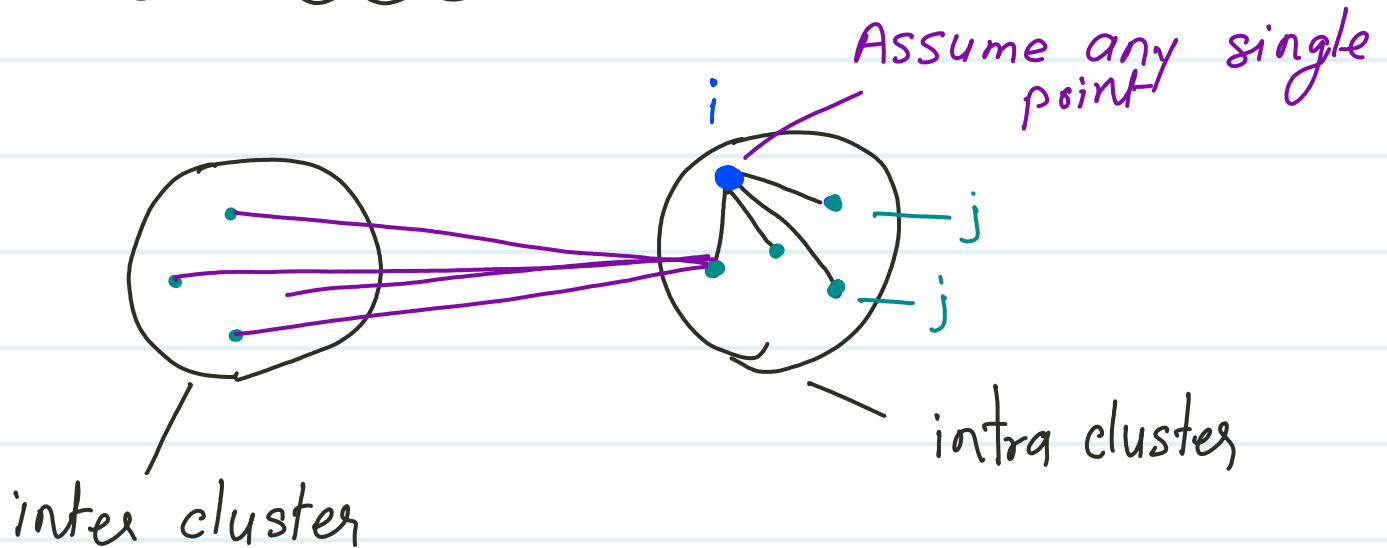




For any clustering method, validation method we will use, that is silhouette score.

Important hyperparameters in the clustering

★ Silhouette score



$$\textcircled{1} \quad a(i) = \frac{\text{Sum of distance of all the point in intra cluster}}{\text{No. of data point}}$$

Formula -

$$a(i) = \frac{1}{|C_i| - 1} \sum_{\substack{j \in C_i \\ i \neq j}} d(i, j)$$

$$(2) \quad b_i = \frac{\text{Distance of all the point of any other cluster}}{\text{no of point}}$$

Formula :-

$$b(i) = \min_{J \neq I} \frac{1}{|C_J|} \sum_{j \in C_J} d(i, j)$$

(3) silhouette formula -

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \text{ if } |C_I| > 1$$

$$s(i) = \begin{cases} 1 - a(i)/b(i), & \text{if } a(i) < b(i) \\ 0, & \text{if } a(i) = b(i) \\ b(i)/a(i) - 1, & \text{if } a(i) > b(i) \end{cases}$$

Range of silhouette score is 1 to -1

1 $\rightarrow$ Good clustering

-1 $\rightarrow$ Bad clustering